OPTT Analysis

Oliver Reece

Abstract

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- What is OPTT? Founded?
- Talk about high risk, share dilution, operating loss
- What is their current market valuation? (penny, market cap)
- Talk about deal with US Gov.
- What do I aim to do originally? What changed? Why?
- What models?
- So what? Looking into whether traditional ML can handle the noise of penny stocks

The Data

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- Where does data come from?
- Time frame?
- Num_observations?
- Key variables?
- Omitted Alt Data

To illustrate the structured dataset, rows 14–19 are displayed below. Because the ATR calculation utilises a 14-day sliding window, the initial 13 observations by default result in NA values. Examining rows 14–19 ensures the inclusion of complete data points.

Year	Month	Day	open_price	close_price	true_range	ATR
2007	5	14	3468	3548	110	223.8571
2007	5	15	3600	3570	150	202.2857
2007	5	16	3500	3456	170	187.2857
2007	5	17	3400	3394	146	174.4286

Year	Month	Day	open_price	close_price	true_range	ATR
2007	5	18	3372	3350	114	157.8571
2007	5	21	3328	3316	126	152.0000

Sentiment Analysis

-placeholder for data table with all 4 articles represented and brief discussion of them-

After querying both NewsAPI and Finnhub, we returned a total of 4 relevant articles. Due to concerns of limited data and subsequent biases, I will not be using sentiment analysis scores within my models.

Instead, the analysis will focus on price-derived technical indicators alongside trading volume. These variables are derived directly from empirical trading data, avoiding the sample-size and bias concerns associated with the limited sentiment data.

In order to maintain structural robustness via alternative data streams, the analysis will pivot to integrate macroeconomic corporate health metrics, specifically metrics such as outstanding share dilution and order backlog growth.

Alternate Data Procurement

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tq_get –> financials –> filter for type == income, unnest quarterly.

problem –> quarterly means 4 data points per year, not a lot of info (data-sparcity bias)

solution –> forward stepwise function to fill for weekly / monthly data (outstanding shares)

solution –> linear interpolation to fill for weekly / monthly data (order backlog growth)